



Running on empty: Diesel-Contaminated sediments impair feeding and behaviour in the marine gastropod *Tritia neritea*

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ABSTRACT

Diesel contamination is a pervasive threat to coastal ecosystems, yet the sublethal behavioural effects on benthic invertebrates remain poorly understood. We exposed the marine gastropod *Tritia neritea* to sediments spiked with two diesel concentrations (DIE-1: 1.4×10^6 ng/g; DIE-2: 3.5×10^6 ng/g) over 28 days to assess survival, foraging, predator-avoidance, burrowing, and biochemical responses. Diesel addition led to initial contamination dominated by heavy hydrocarbons, BTEX compounds, and naphthalene, with DIE-2 exceeding sediment quality guideline thresholds. Hydrocarbon concentrations declined sharply (>96% for C₅–C₄₀, >97% for LMW PAHs) over the experiment, reducing TEQ and MEQ values by > 99%. Despite chemical remediation, snails exposed to DIE-2 exhibited reduced survival ($79 \pm 7.5\%$), impaired foraging success (<25%), prolonged searching times, diminished feeding, and increased immobility. Burrowing activity was slightly enhanced in DIE-2, while righting responses were moderately affected. Biochemical markers related to oxidative stress, neurotransmission and energy reserves remained largely unchanged across treatments. Our results indicate that diesel-contaminated sediments can disrupt feeding and behavioural performance in *T. neritea*, with potential ecological consequences even when contaminant levels decline. These findings highlight the importance of including behavioural endpoints in sediment toxicity assessments.

1. Introduction

Oil pollution in marine ecosystems is an increasingly serious environmental concern (Werorilangi et al., 2020; Kalter and Passow, 2023; Sharma et al., 2024). Numerous anthropogenic activities, including pollutant discharges into rivers, maritime transport, wastewater spills, offshore oil extraction, fuel leaks, and accidental releases, expose aquatic organisms to elevated concentrations of petrochemical compounds (Barbosa et al., 2018; Zapata-Vívenes et al., 2022). Petroleum products enter the ocean in various forms, such as gasoline, diesel, crude oil, and heavy distillates (NRC, 2003; Saeed and Al-Mutairi, 1999), and their toxic effects on both vertebrates and invertebrates are well documented (Bejarano et al., 2006; Müller et al., 2019; Pereira et al., 2018).

Among these pollutants, crude oil and diesel are particularly relevant due to their extensive use in maritime transport (da Silveira Guerreiro et al., 2022). Diesel, the primary fuel for fishing and recreational vessels,

is a major contributor to chronic and acute contamination in coastal waters (Pacheco and Santos, 2001). Mainly composed of polycyclic aromatic hydrocarbons (PAHs) such as phenanthrene, pyrene, and anthracene (Breitwieser et al., 2018), diesel poses risks due to the carcinogenic, mutagenic, and bioaccumulative properties of these compounds. Although diesel exhibits substantial volatility, a portion persists in the marine environment as a surface film (NOAA, 2020), dissolves into the water column (Teles et al., 2003) becoming bioavailable to marine organisms (Kalter and Passow, 2023; Sharma et al., 2024), or binds to suspended particles and sediments, where it may remain for months to decades (Gesteira and Dauvin, 2000; Peterson et al., 2003; Yamamoto et al., 2003). These pathways increase the likelihood of exposure for both pelagic and benthic organisms.

The ecological persistence of petroleum products underscores their significance as global anthropogenic threats to coastal ecosystems (Kottuparambil et al., 2023; Lumibao and Liu, 2024; Uno et al., 2017).

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Despite stringent maritime safety regulations, oil spills remain frequent and are expected to increase as global energy demand rises (Chelomin et al., 2023; Guo et al., 2022). The severity of environmental impacts depends on the type of substance spilled, environmental conditions, and ecosystem vulnerability (de Santana et al., 2021). Pollutants transported by currents and waves often reach intertidal habitats (Kim et al., 2017; Zhou et al., 2019), as occurred during the 2019 Brazil disaster, in which approximately 3000 km of coastline were contaminated by crude oil, severely affecting diverse coastal ecosystems (Soares et al., 2022; de Santana et al., 2021). Despite their ecological importance and high exposure risk, sandy beaches remain one of the least-studied environments in oil spill research (Bejarano and Michel, 2016; Donohoe et al., 2021).

Diesel toxicity has traditionally been investigated through its water-accommodated fraction (WAF), which contains dissolved, suspended, and emulsified hydrocarbons (OECD, 2019) and is considered the most toxic component (da Silveira Guerreiro et al., 2022). Studies on WAF toxicity have demonstrated a wide range of adverse effects on aquatic organisms, including mortality (de Santana et al., 2021; Monteiro et al., 2018), oxidative stress (Nogueira et al., 2013; Milinkovitch et al., 2015; Feijó de Oliveira et al., 2015; Sandrini-Neto et al., 2016), growth inhibition (Sagerup et al., 2016; Pereira et al., 2018; Brown et al., 2020; Werorilangi et al., 2020; de Santana et al., 2021; Donohoe et al., 2021), and reproductive alterations (Han et al., 2017; de Santana et al., 2021). Additional impacts include changes in microbial diversity (Durand et al., 2023), transcriptomic responses (Imbery et al., 2024), DNA damage (Delunardo et al., 2019), cytotoxicity (Katsumiti et al., 2019), and immune alterations in invertebrates (Zapata-Vivenes et al., 2022). In contrast, the effects of diesel-contaminated sediments have received far less attention (Won et al., 2018; Ghosh and Mandal, 2021), despite sediment acting as a long-term reservoir and a key exposure route for benthic organisms.

Behavioural responses are increasingly recognized as sensitive indicators of sublethal toxicity yet remain underexplored in hydrocarbon-related ecotoxicology (Bejarano et al., 2006). Behavioural alterations often occur at concentrations lower than those producing biochemical or lethal effects, as observed in the amphipod *Paramoera walkeri* (Brown et al., 2017). In molluscs, behavioural impairments have been reported following exposure to crude oil (Stekoll et al., 1980), diesel WAF (Nagarajah et al., 1985; Werorilangi et al., 2020; da Silveira Guerreiro et al., 2022), and diesel-contaminated sediments (Chapman et al., 1988). Such disruptions may reduce predator avoidance, impair feeding efficiency, and compromise reproduction, with potential consequences at both population and ecosystem scales (van Wezel and Opperhuizen, 1995; Seuron, 2011; Brown et al., 2017).

Given the persistence of diesel in coastal sediments, there is a critical need to evaluate how sediment-bound hydrocarbons affect benthic invertebrates using ecologically relevant endpoints. In this study, we investigated the behavioural responses of the marine snail *Tritia neritea* (Linnaeus, 1758) to diesel-contaminated sediments, testing the null hypothesis that exposure does not affect survival, predator avoidance, burrowing, foraging, or biochemical parameters. By integrating these multi-level responses, this research underscores the significance of behavioural endpoints as early-warning signals. Our findings demonstrate how subtle alterations in locomotion and foraging success provide essential insights into the broader ecological consequences that propagate through benthic food webs.

2. Materials and methods

2.1. Target species

Tritia neritea (Linnaeus, 1758) is a scavenger/detritivore inhabiting sandy and muddy intertidal zones across the Black Sea, Mediterranean and Atlantic (Boissin et al., 2020; Marín Rodríguez et al., 2021). As a burrower, it contributes to nutrient and energy transfer on sandy

beaches (Zhang et al., 2021). Like most gastropods, it uses chemoreceptors to detect food (Ansell, 2001; Morton and Yuen, 2000) and predators (Jacobsen and Stabell, 1999). It remains buried during low tide (Morton, 1960), emerging rapidly when detecting carrion (Brown, 2001), which it consumes quickly to reduce predation risk (McKillup and McKillup, 1994; Morton et al., 1995) before avoiding desiccation by returning to the intertidal zone (McLachlan et al., 1979). Because of its close association with sediment, *T. neritea* is highly exposed to substrate-bound pollutants, with ecological implications for its predators (Baeta and Ramón, 2013; Sprung, 1993).

2.2. Fuel and sediment

Stock sediment was prepared by mixing 50 mL diesel (Repsol e+) with 500 g fine sand (1% organic matter). After 48 h of agitation and overnight settling, aliquots were sent to INTERTEK (<https://www.intertek.com>) for quantification of Σ TPH and the 16 USEPA priority PAHs (US-EPA, 1984; 2002). Two treatments were prepared from this stock: 10% (DIE-1) and 25% (DIE-2). These exposure levels were selected to reflect realistic contamination scenarios (e.g., Ghandourah, 2022; Ofori et al., 2020; Teixeira et al., 2024) and established sediment quality guidelines (NOAA, 1999). This approach simulates different spill magnitudes to assess their impact over time. Furthermore, it allows for the study of recovery processes mediated by abiotic factors, including volatilization and hydrodynamic flushing of the sediment. Therefore, sediments were reanalysed at the end of the experiment to determine changes after periodic seawater renewal simulating natural hydrodynamic washing (Kim et al., 2017; Zhou et al., 2019). Concentrations of low- and high-molecular-weight PAHs were determined, as well as Σ cPAHs (Wang et al., 2016). Toxicity was assessed using two sediment quality guidelines (Effect Range Low-ERL and Effect Range Median-ERM: Long et al., 1995; Threshold Effect Level-TEL and Probable Effect Level-PEL: Macdonald et al., 1996), distinguishing ranges of low, possible, or probable effects (Adeleye et al., 2016). The toxic equivalents (TEQ) and carcinogenic toxic equivalency factor (TEF_{car}) were calculated using toxic and mutagenic equivalency factors relative to BaP (Dong et al., 2014; Nadal et al., 2004; Tsai et al., 2004). The carcinogenic toxic equivalent quotient (TEQ) and mutagenic equivalent quotient (MEQ) of the PAH mixture were calculated according to Eqs. (1) and (2):

$$\text{TEQ} = \sum \text{Cn} \times \text{TEF} \quad (1)$$

$$\text{MEQ} = \sum \text{Cn} \times \text{MEF} \quad (2)$$

where Cn is the concentration of each PAH, and TEF and MEF are the toxic and mutagenic equivalency factors relative to benzo(a)pyrene (BaP) (Nekahavhambe et al., 2014; Adeniji et al., 2019).

2.3. Sampling and acclimation

Snails were hand-collected during low tide in the Bay of Cádiz and transported under aeration. In the laboratory (7 days of acclimation), they were maintained in a 20 L aquarium with fine natural colourless sand (0.2–0.6 mm, suitable for burrowing species; JBL Sansibar Grey) and artificial seawater (34 ± 1), under controlled photoperiod, temperature (20 ± 1 °C), pH (8.1), oxygen (>85%), and 50% water renewal every 3 days. They were fed with *Solen marginatus* and *Mytilus galloprovincialis* twice weekly (García-García et al., 2015; Gonçalves et al., 2017). Individuals of similar size (4.33 ± 0.31 mm) were selected to minimize size-dependent variability.

2.4. Experimental design

Ten similarly sized snails per treatment (control, DIE-1, DIE-2; n = 3 replicates) were exposed for 28 days. Survival was recorded weekly.

Behavioural assays (self-righting, trophic interactions, predation susceptibility) were performed on days 7, 14 and 28; substrate selection was tested on day 21 (Fig. 1). In each assay, individuals were used once and isolated post-testing. After testing, snails were fed and burrowing success was recorded within 5 min upon their return to their aquaria.

Self-righting was measured following Manríquez et al. (2014, 2019). Snails were placed upside down on a roughened surface inside a Petri dish with 0.5 cm ASW, with the shell aperture positioned parallel to the dish surface to standardize difficulty. A 3×3 cm piece of rough-textured rubber (3 M Safety-Walk™ tape) was fixed to the bottom of the dish to aid adhesion, on which the snail was placed. Maximum time allowed was 7 min; failure to right was recorded as unsuccessful. Activity state (immobile/attempting) and number of attempts were noted. Each snail was used only once, with 10 replicates per treatment (2–3 individuals per aquarium).

Trophic interaction test. After 7 days of fasting (Garton and Stickle, 1980; Giacoletti et al., 2017), a Y-maze two-choice feeding assay was conducted following Xu et al. (2017) and Marín-Rodríguez et al. (2021) using two pieces of carrion (0.33 g each), from the same food species used during acclimation. Four Y-mazes were run simultaneously, and behaviour was recorded (Sony Handycam HDR-CX240E) for 15 min (Morton and Jones, 2003; Aloy et al., 2011). Ten individuals per treatment were tested at days 7 and 28. Measured variables included foraging success (proportion of successful individuals), searching and feeding

times (time to reach carrion and from first contact to food abandonment), behavioural states (stationary-1, moving without feeding-2, searching for food-3, and prey-handling or feeding-4) (Hu et al., 2016; Sun et al., 2016), Chesson's selectivity index (Chesson, 1983) and food intake (Marín Rodríguez et al., 2021). After each trial, aquaria were cleaned to remove chemical cues or residual that might affect subsequent tests. Failure to reach food within 15 min was classified as unsuccessful.

Substrate selection. After 21 days, habitat selection across diesel-contaminated sediments was tested using a modified multiple-choice design (Araújo et al., 2012, 2014, 2016a, 2016b) to evaluate ecological consequences related to emigration, habitat selection, and (re) colonization (De Lange et al., 2006; Moe et al., 2013). To avoid leader-following behaviour due to mucus trails (Barajas-Ledesma and Holland, 2023; Manríquez et al., 2014; Newar et al., 2021), trials used single individuals. Five plates, each containing 12 compartments (with 22.09 cm^2 and 75 g sediment per compartment), were used. We included 4 replicates for each sediment type, arranged randomly using an algorithm (Fig. 2). Movements were recorded for 45 min. Parameters included substrate chosen, time to choice, onset of burrowing, and total burrowing time. At the end of the exposure period, snails were frozen in liquid nitrogen. Soft tissues were homogenized under liquid nitrogen and divided into 0.2 g fresh weight (FW) aliquots.

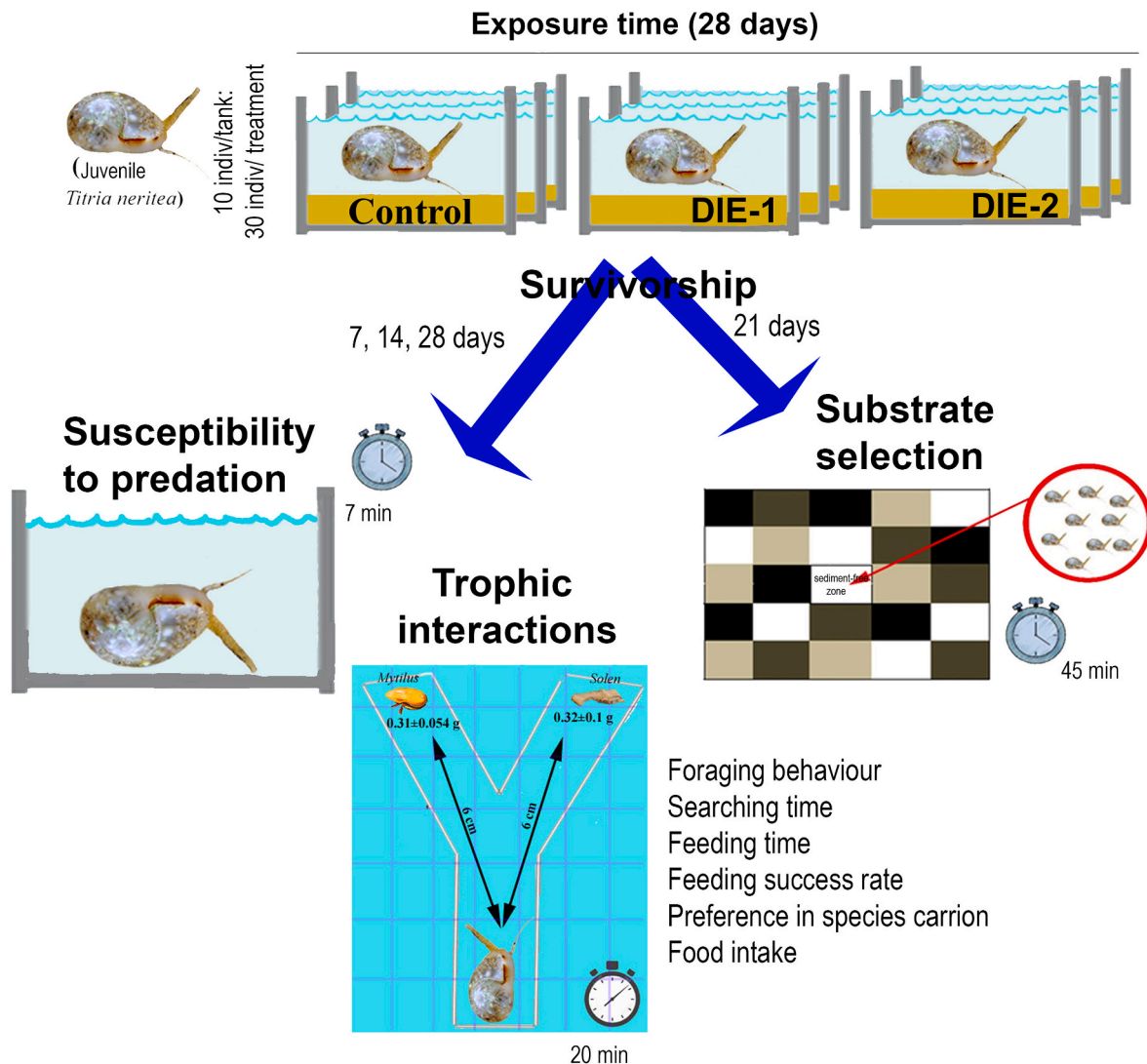


Fig. 1. Diagram of the experimental design. Diesel treatments: Control = 0, DIE-1 = 10%, DIE-2 = 25%.

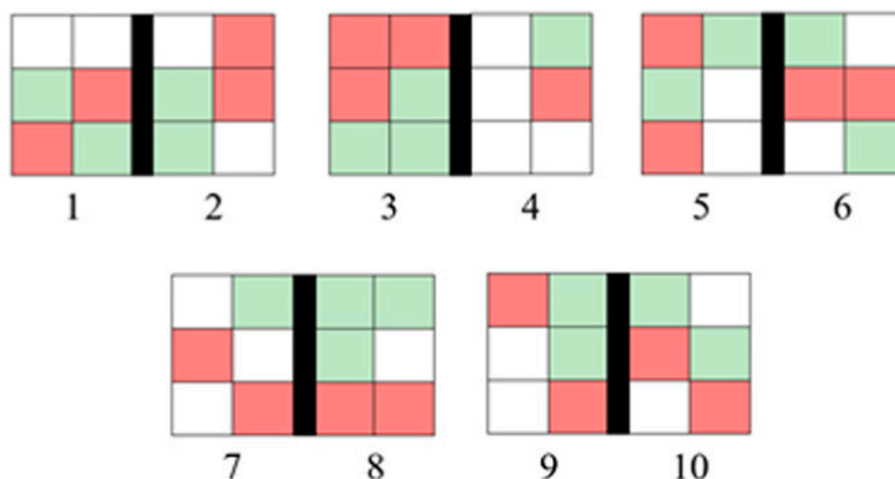


Fig. 2. Schematic representation of the random distribution of sediment types on each plate. White indicates the control treatment; green, the DIE-1 treatment; and red, the DIE-2 treatment. The black area represents the substrate-free zone where the snail was placed. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.5. Biochemical analysis

Biochemical responses to diesel exposure were evaluated using biomarkers of: (i) metabolic reserves (glycogen content, GLY), (ii) antioxidant defence (superoxide dismutase activity, SOD), (iii) detoxification capacity (carboxylesterase activity, CbEs), (iv) cellular damage (lipid peroxidation levels, LPO), and (v) neurotoxicity (acetylcholinesterase activity, AChE). Each biomarker was analysed in 3 samples per treatment (1 individual per aquarium; 3 aquaria per treatment), including technical duplicates.

Tissue homogenization was performed at a 1:2 (w/v) ratio in the appropriate extraction buffer using a TissueLyzer II (Qiagen; 20 Hz, 90 s). For GLY, SOD, CbEs, and AChE, tissues were extracted with 50 mM potassium phosphate buffer and centrifuged at 10,000 g for 20 min at 4 °C. LPO analysis used 20% trichloroacetic acid (TCA) as extraction buffer, followed by centrifugation under the same conditions. Supernatants were either processed immediately or stored at −80 °C.

Energy reserves: GLY concentration was determined using the phenol-sulfuric acid method (DuBois et al., 1956). Absorbance was measured at 492 nm, with glucose standards (0–5 mg/mL). Results were expressed as mg/g FW.

Antioxidant defences: SOD activity was measured following Magnani et al. (2000), based on inhibition of pyrogallol. Absorbance was read at 420 nm after 1 min of incubation. Activities were expressed as U (an inhibition of 50 % of pyrogallol autooxidation)/g FW.

Biotransformation enzymes: CbEs activity was quantified according to Hosokawa and Satoh (2001), modified by Solé et al. (2018), using p-nitrophenyl acetate (pNPA) and p-nitrophenyl butyrate (pNPB) as substrates. Absorbance at 405 nm was recorded every 15 s for 5 min, using $\epsilon = 18 \text{ mM}^{-1} \text{ cm}^{-1}$. Results were expressed as nmol/min/g FW.

Cellular damage: LPO levels were determined following Ohkawa et al. (1979), modified by Carregosa et al. (2014), quantifying malondialdehyde (MDA) via the thiobarbituric acid reactive substances (TBARS) assay. Absorbance was read at 532 nm using $\epsilon = 156 \text{ mM}^{-1} \text{ cm}^{-1}$. Results were expressed as nmol MDA/g FW.

Neurotoxicity: AChE activity was measured following Ellman et al. (1961), with modifications by Mennillo et al. (2017), using acetylthiocholine iodide (ATChI). Absorbance at 412 nm was recorded over 5 min (25 s intervals) using $\epsilon = 13.6 \text{ mM}^{-1} \text{ cm}^{-1}$. Activity was expressed as nmol/min/g FW.

2.6. Data analysis

The null hypothesis—that diesel contamination does not affect survival, behaviour, or biochemistry of *Tritia neritea* over time—was tested using repeated-measures ANOVA (RM-ANOVA). Time (Ti; 7, 14, and 28 days for all endpoints, with an additional 21-day sampling point for survival analysis) was the within-subjects factor, and diesel treatment (DIE: control, DIE-1, DIE-2) was the between-subjects factor. Pillai's trace was used for multivariate effects; Greenhouse–Geisser corrections were applied when sphericity was violated. Tukey's post-hoc test was used for between-subjects comparisons and Bonferroni corrections for within-subjects effects. Significant interactions were further explored with Bonferroni tests.

The percentage of food-searching success, time to substrate selection, substrate chosen, and biochemical biomarkers were analysed using one-way ANOVA. A two-way ANOVA was used for behavioural states and Chesson's selectivity index. Normality and homoscedasticity were assessed using the Kolmogorov–Smirnov and Levene's tests. Chi-square tests were applied to assess differences in prey-choice frequencies and whether proximity influenced substrate selection. Statistical significance was set at $p < 0.05$ for all analyses. All analyses were performed in IBM SPSS Statistics v21.

3. Results

3.1. Substrate analysis

As expected, diesel addition to pristine sediments resulted in contamination dominated by heavy hydrocarbons ($\text{C}_{10}\text{--C}_{40}$). Among BTEX (benzene, toluene, ethylbenzene, and xylenes) compounds, m,p-xylene and σ -xylene were the most abundant, reaching 1.6 and 1.1 mg/kg in DIE-1 and 4.0 and 2.7 mg/kg in DIE-2, respectively. Together, these compounds accounted for 45.3% and 30.6% of total BTEX, clearly exceeding benzene levels (Table S1). Naphthalene (NAP), the predominant two-ring PAH, reached 0.58 mg/kg in DIE-1 and 1.45 mg/kg in DIE-2 (36.3% of $\sum\text{PAH}_{16}$). Among HMW PAHs, pyrene (PYR, four rings) was dominant, ranging from 0.44 to 1.10 mg/kg (27.8% of $\sum\text{PAH}_{16}$). Although both sediments exhibited similar chemical profiles, hydrocarbon concentrations in DIE-2 were approximately 2.5 times higher across all compound groups.

At the end of the experiment, a marked reduction in hydrocarbon concentrations was observed. The $\text{C}_5\text{--C}_{40}$ fraction decreased by ~96–97%, and BTEX compounds were almost completely removed

(>99%). LMW PAHs (2–3 rings), including NAP, acenaphthylene (ACY) and fluorene (FLR), declined by 97–99.8%, with fluoranthene (FLT) being the most persistent. Despite the ~99% reduction, NAP remained the most abundant PAH (DIE-1: 0.0016 mg/kg; DIE-2: 0.013 mg/kg), contributing 9.4–36.3% to Σ PAH₁₆. In contrast, the relative abundance of HMW PAHs (≥ 4 rings) increased substantially in both sediments (>70%). PYR remained the most abundant of these compounds, ranging from 0.011 mg/kg in DIE-1 (64.6% of Σ PAH₁₆) to 0.0207 mg/kg in DIE-2 (58.6% of Σ PAH₁₆).

According to sediment quality guidelines (ERL/ERM and TEL/PEL), initial PAH contamination and associated ecological risks differed between sediments (Table S2). In DIE-2, NAP, fluorene, acenaphthene and acenaphthylene exceeded TEL values, and FLR surpassed the ERL threshold. In contrast, DIE-1 showed only low to moderate contamination, with NAP and dibenz (a,h)anthracene approaching TEL limits. By the end of exposure, the strong decline in PAHs resulted in all compounds falling below TEL and ERL thresholds. Based on carcinogenic toxic equivalent quotient (TEQ) and mutagenic equivalent quotient (MEQ), both sediments initially exhibited TEQ and MEQ values of 23.23–23.58 and 18.24–19.62 μ g BaP-eq/g, respectively, which decreased by > 99% after 28 days (0.031–0.061 μ g BaP-eq/g). DIE-2 retained slightly higher values due to the persistence of benzo(a)anthracene, suggesting differential degradation and retention of less bioavailable aromatic fractions (Table S3).

3.2. Behavioural responses

Survival and burrowing: Diesel exposure significantly affected *T. neritea* survival (Fig. 3), although no significant effects of exposure time or treatment–time interaction were detected (Table 1). Survival was 100% in Control and DIE-1 treatments, while DIE-2 exhibited significantly lower survival ($79 \pm 7.5\%$).

Burrowing success was also influenced by diesel exposure, although no interaction with time was detected (Table 1). Five minutes after placement on the sediment, snails in the DIE-2 treatment exhibited significantly higher burrowing success ($75.1 \pm 19.1\%$) than those in DIE-1 and Control.

Predation susceptibility (self-righting): No significant differences in righting success were observed among treatments (Table 2), although DIE-2 snails exhibited a lower success rate (74%) than Control and DIE-1

Table 1

Repeated measures ANOVA results for survival and burrowing behaviour of *Tritia neritea* under different diesel treatments (Control, DIE-1, DIE-2) over time. Mean square (MS) and degrees of freedom (df) are shown.

		df	MS	F	p
Survival	Time	1.105	142.841	3.641	0.099
	Time*Treatment	2.210	142.841	3.641	0.083
	Treatment	2	1613.361	136.021	< 0.0001
	Error (time)	6.531	39.235		
	Error (treatment)	6	11.861		
Burrowing success	Time	1.614	1724.378	3.333	0.086
	Time*Treatment	3.228	1704.357	3.295	0.660
	Treatment	2	1581.463	686.673	0.012
	Error (time)	9.683	517.324		
	Error (treatment)	6	156.925		

(>85%). A significant interaction between treatment and exposure time was detected, as Control and DIE-1 behaved differently over time compared with DIE-2. Righting time and the number of righting attempts did not differ among treatments, and no treatment–time interactions were found (Table 2).

Trophic interactions: Foraging success differed significantly among treatments (df = 2, MS = 2749.648; F = 10.353; p = 0.011). Control and DIE-1 snails achieved success rates $\geq 75\%$, forming a homogeneous group (Tukey's test), whereas DIE-2 individuals did not exceed 25% (Fig. 4). Searching time followed the same trend: DIE-2 snails took nearly twice as long to locate food compared to individuals in the other treatments. Exposure time did not significantly influence this parameter, and no interaction with treatment was detected (Table 3).

Feeding time differed across treatments. After 7 days, Control individuals spent twice as long feeding as diesel-exposed snails. At 14 and 28 days, feeding time was similar in Control and DIE-1, but significantly lower in DIE-2 (<106 s), with only a few individuals able to locate and consume prey.

Food intake was similar in Control (0.0356 ± 0.032 g) and DIE-1 (0.042 ± 0.0635 g) treatments, regardless of carrion species. In contrast, DIE-2 snails consumed significantly less food (0.0083 ± 0.0176 g). Exposure time had no significant effect, and no interaction

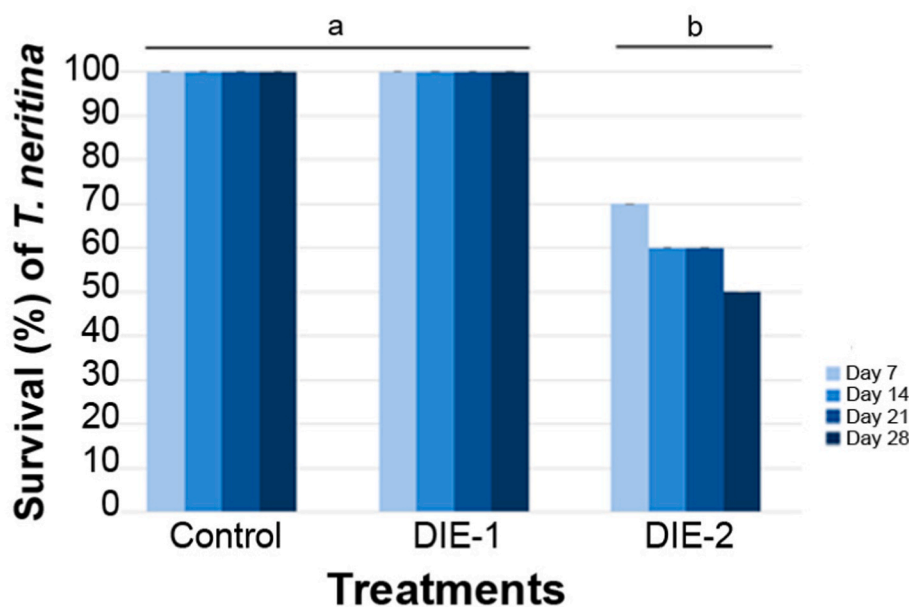


Fig. 3. Survival (%) of *Tritia neritea* under different diesel treatments (Control = 0, DIE-1 = 10%, DIE-2 = 25%) at different exposure times (7, 14, 21, and 28 days). Mean values $\pm$ SD. Significant differences between groups are indicated by different lowercase letters.

Table 2

Repeated measures ANOVA results for predation susceptibility (self-righting success, time, and number of righting attempts) of *Tritia neritea* under different diesel treatments (Control, DIE-1, DIE-2) over time. Mean square (MS) and degrees of freedom (df) are shown.

		df	MS	F	p
Self-righting success	Time	1.653	2143.591	4.521	0.055
	Time*Treatment	3.306	423.540	0.893	0.486
	Treatment	2	1154.651	3.502	0.098
	Error (time)	9.918	474.050		
	Error (treatment)	6	329.671		
Self-righting time	Time	1.911	94530.973	5.660	0.070
	Time*Treatment	3.823	22388.041	1.341	0.268
	Treatment	2	60828.078	1.867	0.174
	Error (time)	51.608	16700.348		
	Error (treatment)	27	32573.815		
Number of righting attempts	Time	1	0.817	0.272	0.606
	Time*Treatment	2	2.817	0.938	0.404
	Treatment	2	3.317	1.775	0.189
	Error (time)	27	3.002		
	Error (treatment)	27	1.869		

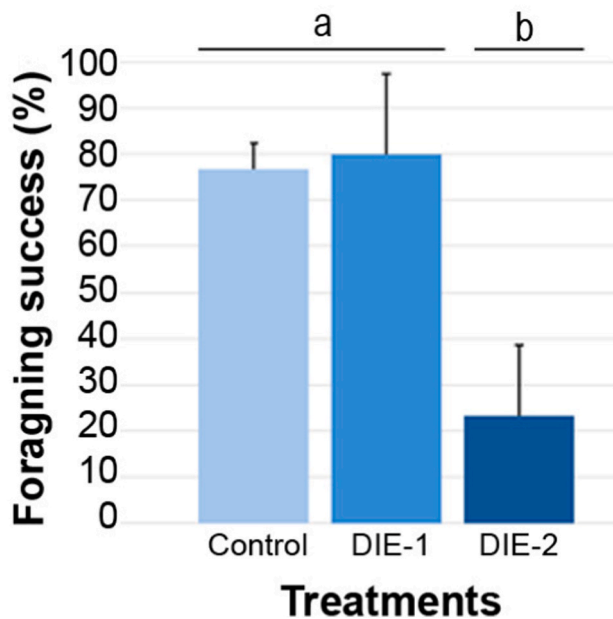


Fig. 4. Foraging success (%) of *Tritia neritea* under different diesel treatments (Control = 0, DIE-1 = 10%, DIE-2 = 25%). Mean values $\pm$ SD. Significant differences between groups are indicated by different lowercase letters.

between treatment and time was detected (Table 3).

Foraging behaviour differed significantly among treatments (Table 4). Snails spent about 50% of their time moving without searching (state 2). In Control and DIE-1, ~25% of the time was devoted to active searching (state 3), whereas in DIE-2 this proportion was halved. DIE-2 snails also spent more time motionless (30%, state 1) and did not exhibit feeding behaviour (state 4), unlike Control and DIE-1 individuals, which spent ~10% of their time feeding. A significant treatment $\times$ behaviour interaction supported these differences (Table 4).

The Chesson index showed no significant differences among treatments, indicating a general preference for *Solen marginatus* over *Mytilus galloprovincialis*. No treatment $\times$ prey interaction was detected. However, ANOVA revealed significant differences in prey selection (Table 4). Pearson's chi-square tests indicated that on day 7, all treatments

Table 3

Repeated measures ANOVA results for foraging activity (searching time and feeding time) and food intake of *Tritia neritea* under different diesel treatments (Control, DIE-1, DIE-2) over time. Mean square (MS) and degrees of freedom (df) are shown.

		df	MS	F	p
Searching time	Time	1.920	223926.133	1.177	0.315
	Time*Treatment	3.840	113476.906	0.597	0.660
	Treatment	2	2886047.24	21.73	< 0.0001
	Error (time)	51.838	190221.882		
	Error (treatment)	27	132831.669		
Feeding time	Time	2	140472.844	0.541	0.585
	Time*Treatment	4	967053.694	3.728	0.009
	Treatment	2	3150796.68	13.246	< 0.0001
	Error (time)	54	259415.609		
	Error (treatment)	27	237870.653		
Food intake	Time	2	0.000	0.235	0.792
	Time*Treatment	4	0.003	1.680	0.168
	Treatment	2	0.009	6.191	0.006
	Error (time)	54	0.002		
	Error (treatment)	27	0.002		

Table 4

ANOVA results for foraging behaviour and Chesson's selectivity index for carrion preference in *Tritia neritea* under different diesel treatments (Control, DIE-1, DIE-2) over time. Mean square (MS) and degrees of freedom (df) are shown.

		df	MS	F	p
Foraging behaviour	States	3	0.314	282	< 0.0001
	States*Treatment	6	0.013	11.417	< 0.0001
	Error	24	0.001		
Chesson selectivity index	Treatment	2	0.000	0.000	1
	Carrion	1	0.857	19.146	0.001
	Treatment*Carrion	2	0.005	0.106	0.900
	Error	12	0.045		

preferred razor clams. By days 14 and 28, prey selection patterns diverged: Control snails continued to prefer razor clams (71.43%) but also consumed mussels, whereas diesel-exposed snails selected only razor clams on day 14 (with only one DIE-2 individual feeding). By day 28, both DIE-1 and DIE-2 snails consumed *Mytilus*, with frequencies of 50% and 25%, respectively.

In terms of trophic interactions, significant differences were observed in foraging success (df = 2, MS = 2749,648; F = 10.353, p = 0.011). Snails in the Control and DIE-1 treatments achieved average foraging success rates $\geq$ 75%, whereas those in the DIE-2 treatment did not exceed 25%, significantly differing from the homogeneous group formed by the Control and DIE-1 treatments (Tukey's test, p < 0.05) (Fig. 4). A similar trend was observed for searching time, with DIE-2 snails taking nearly twice as long to locate food compared to individuals in the other two treatments. However, exposure time did not significantly affect this parameter, and no interaction between treatment and exposure time was detected (Table 3).

Regarding feeding time, control individuals spent twice as long feeding as those in the diesel treatments after 7 days of exposure. However, at 14 and 28 days no differences were observed between the feeding time of Control and DIE-1 snails, although significant differences were found between these groups and DIE-2. In the latter, feeding time was less than 106 s, and only a few individuals managed to locate and consume prey.

No significant differences in the Chesson index were detected among

treatments, indicating a general preference of *T. neritea* for the native bivalve *S. marginatus* over *M. galloprovincialis*, regardless of the treatment. Additionally, no significant interaction between treatment and prey type was observed. However, ANOVA revealed significant differences in prey selection (Table 4). Pearson's chi-square analysis indicated that on day 7, individuals from all treatments preferred razor clams. By days 14 and 28, significant differences in prey selection frequencies emerged. Control snails continued to prefer razor clams (71.43%) but also included mussels in their diet. In contrast, diesel-exposed snails exclusively selected razor clams on day 14, with only one DIE-2 individual feeding. By day 28, both DIE-1 and DIE-2 snails began to consume *Mytilus*, with frequencies of 50% and 25%, respectively.

Substrate selection: In the Control, 70% of snails selected and burrowed into a suitable substrate. In diesel-exposed groups, burrowing decreased to 60%, with one (DIE-1) and three (DIE-2) individuals remaining unburied. DIE-2 also showed a slight increase (20%) in immobile individuals. However, no significant differences in substrate

selection were found among treatments ($df = 2$, $MS = 70.001$; $F = 0.159$; $p = 0.856$), nor was there preference for uncontaminated sediment ($df = 2$, $MS = 840.444$; $F = 0.846$; $p = 0.475$) or for the sediment previously experienced during exposure ($df = 2$, $MS = 329.333$; $F = 0.244$; $p = 0.791$). According to the chi-square test, substrate selection was not proximity-dependent ($\chi^2 = 0.591$; $p = 0.485$).

3.3. Biochemical responses

Snails in the Control group had slightly higher glycogen reserves than diesel-exposed individuals, although differences were not significant ($df = 2$; $MS = 2.332$; $F = 0.669$; $p = 0.547$). Diesel exposure did not induce antioxidant (SOD; $df = 2$; $MS = 0.91$; $F = 1.526$; $p = 0.291$) or biotransformation (CbEs; $df = 2$; $MS = 959.425$; $F = 0.213$; $p = 0.814$) responses, although CbEs activity was slightly higher in DIE-1. LPO levels were similar across treatments ($df = 2$; $MS = 0.107$; $F = 0.040$; $p = 0.961$). AChE activity was elevated in DIE-1 but did not differ

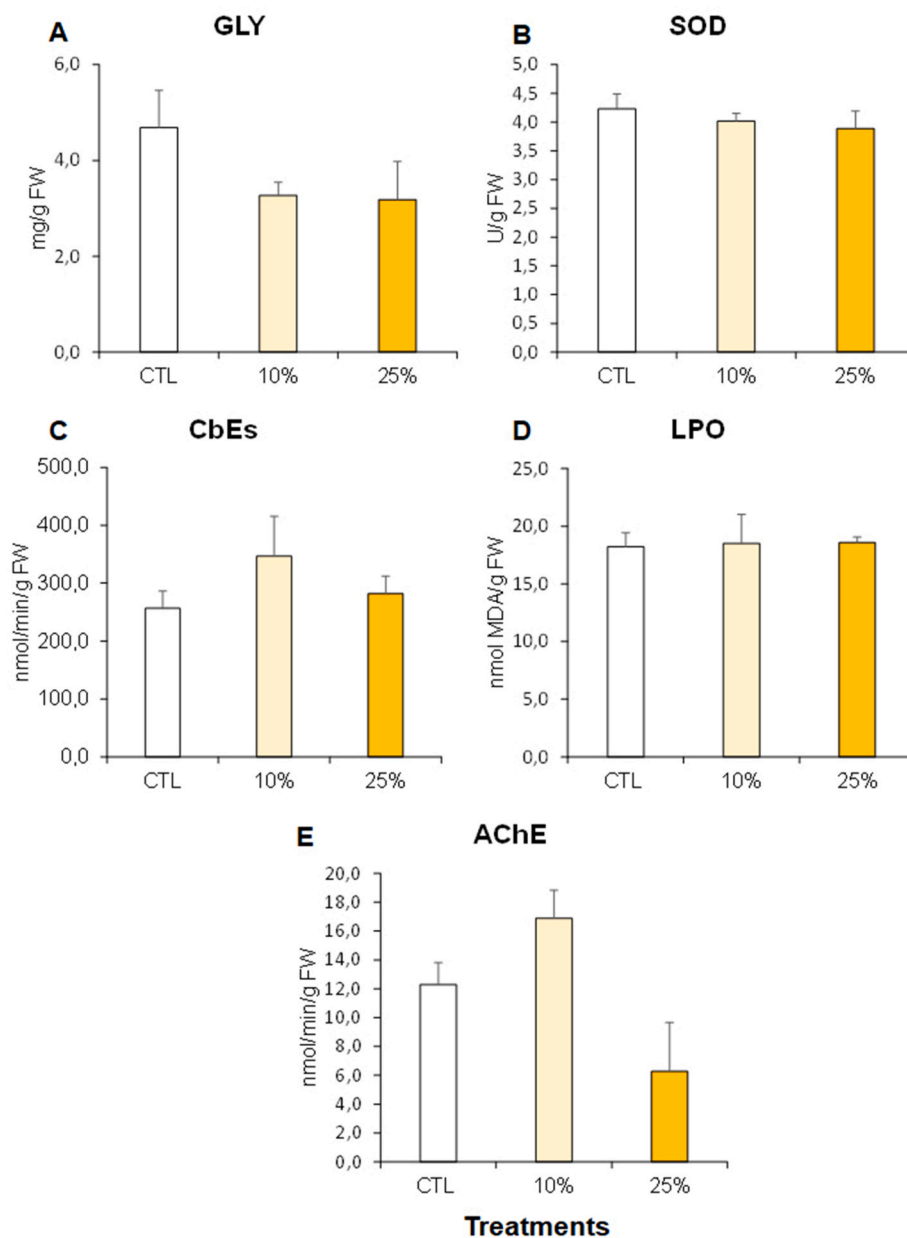


Fig. 5. A) Glycogen content (GLY); B) superoxide dismutase activity (SOD); C) carboxylesterase activity (CbEs); D) lipid peroxidation levels (LPO); and E) acetylcholinesterase activity (AChE) in *Tritia neritea* exposed to different diesel concentrations (Control = 0, DIE-1 = 10%, DIE-2 = 25%). Mean values $\pm$ SD.

significantly among treatments ($df = 2$; $MS = 36.278$; $F = 1.974$; $p = 0.219$) (Fig. 5).

As a summary, Table 5 concisely presents the results obtained from the behavioural and biochemical responses of *Tritia neritea* to diesel exposure.

4. Discussion

Marine ecosystems are increasingly exposed to total petroleum hydrocarbons (TPHs) originating from offshore oil extraction, transport, accidental spills, fossil fuel combustion, riverine discharges, shipping activities, and sewage effluents (Li et al., 2019; Meng et al., 2019). Most studies have focused on acute and chronic toxicity endpoints—survival, growth, and reproduction (Sagerup et al., 2016; Han et al., 2017; Monteiro et al., 2018; Pereira et al., 2018; Brown et al., 2020; Werorilangi et al., 2020; de Santana et al., 2021; Donohoe et al., 2021)—leaving a significant gap concerning the sublethal behavioural effects of TPHs on marine fauna. Behavioural responses such as foraging or predator avoidance are highly sensitive indicators of exposure because they directly influence ecological fitness and the ability to respond to environmental cues (Ford et al., 2021). Given the neuroendocrine-disrupting potential of several TPH components, particularly polycyclic aromatic hydrocarbons (PAHs), evaluating early and long-term behavioural alterations is crucial for understanding impacts in vulnerable coastal ecosystems (Collins et al., 1998; Devi et al., 2016; Kuo et al., 1998; Rengarajan et al., 2015; Yifei et al., 2024).

In this study, sediments were spiked with diesel to achieve initial TPH concentrations of 1.4×10^6 ng/g (DIE-1) and 3.5×10^6 ng/g (DIE-2), corresponding to PAH levels of 1.6×10^6 and 4.0×10^6 ng/g, respectively. Total PAH concentrations in aquatic sediments vary widely (7.0×10^4 to 1.0×10^9 ng/g), with the highest values in urban or industrialised coastal (Du and Jing, 2018; Ghandourah, 2022; Maletic et al., 2019; Ofori et al., 2020; Teixeira et al., 2024). Along the Galician coast—affected by multiple oil spills, including the Prestige-PAH concentrations ranged from 25 to 4000 ng/g (Monaco et al., 2017). Based on TPH, the experimental sediments correspond to “contaminated” (DIE-1) and “severely contaminated” (DIE-2) categories, whereas PAH levels fall within NOAA's (1999) low- (DIE-1) and medium-effect (DIE-2) ranges. Naphthalene (NAP) and pyrene (PYR) dominated both treatments, contrasting with the fluoranthene–pyrene dominance typical of Galician sediments (Monaco et al., 2017).

After 28 days, the C_{5-40} fraction declined by 96–97%. BTEX compounds were almost fully removed (>99%), and LMW PAHs showed marked depletion, generating a relative increase in HMW PAH proportions. This reflects the high hydrophobicity and partition coefficients of HMW PAHs (Sheng et al., 2021), and the greater volatility and

degradability of LMW PAHs (Yu et al., 2019; Miao et al., 2023). Under natural conditions, seawater-leachable TPH fractions are dominated by volatile LMW PAHs that evaporate or degrade rapidly (Zhao and Pignatello, 2004; Wang et al., 2022), whereas medium- and HMW PAHs persist in sediments, particularly those enriched in organic matter (Baumard et al., 1999; La Guardia et al., 2012).

PAH mobilisation during sediment resuspension results from both sedimentary processes (resuspension, aggregation–dispersion) and physicochemical interactions (adsorption–desorption, dissolution–precipitation) (Latimer et al., 1999; Feng et al., 2008; Dong et al., 2016; Vagge et al., 2018). Waves or hydrodynamic energy can resuspend contaminated sediments, facilitating PAH transport (Edge et al., 2014; Labianca, 2021; Lu et al., 2025), and similar dynamics can occur during water renewal in aquaria (Kim et al., 2017; Zhou et al., 2019). The low HMW PAH concentrations observed here likely arise from the coarse sediment used (rounded sand, 0.36 mm mean grain size). PAH levels typically correlate negatively with sand content (Guo et al., 2009; Buckley and Winters, 2011; Saria-Villa et al., 2016; El-Maradny et al., 2023) and positively with organic carbon (Barakat et al., 2011; Bergamasco et al., 2014; Hatzianestis et al., 2020; Ambade et al., 2021; El-Maradny et al., 2023). Although 1% organic matter was added, it was not finely particulate and may have been lost during water exchange, along with associated PAHs—consistent with the fact that >90% of PAHs in sediments are bound to particulate organic matter (Das et al., 2025). Bioturbation may also contribute to the redistribution or removal of hydrocarbons, as observed in burrowing macrofauna (Wenbo et al., 2021).

The distribution of LMW and HMW PAHs provides insight into their toxicity (Ambade et al., 2022; Eça et al., 2021). Huang et al. (2023) found strong correlations between sediment TEQc and Σ_{16} PAH concentrations, especially HMW PAHs, because the highest toxic equivalency factors (TEFs) correspond to HMW compounds. Thus, TEQ–PAH correlations can support rapid toxicity assessments. At the beginning of the experiment, TEQ and MEQ values indicated considerable carcinogenic and mutagenic potential, driven mainly by HMW PAHs such as benzo(a)pyrene, dibenz (a,h)anthracene, and benzo(b)fluoranthene, which exceeded ecological risk thresholds (≥ 0.1 μ g BaP-eq/g) and may pose genotoxic risks to benthic fauna (Nisbet and LaGoy, 1992; Durant et al., 1996, 1999). However, the >99% reduction in carcinogenic and mutagenic PAHs by the end of the experiment lowered TEQ and MEQ below recognized risk limits.

Initially, DIE-2 sediments showed moderate contamination by LMW PAHs—particularly naphthalene, fluorene, acenaphthene, and acenaphthylene—with fluorene exceeding its ERL threshold. This suggests a moderate likelihood of adverse effects on *Tritia neritea*. DIE-1 sediments showed only low to moderate contamination. By the end of exposure, all PAH were below TEL and ERL guideline values, indicating negligible ecological risk (Long et al., 1995; MacDonald et al., 1996).

Individual PAHs may not elicit effects equivalent to the sum of their constituent compounds (Di Toro et al., 2000; de Santana et al., 2021; Albarano et al., 2023). Additive or antagonistic interactions and matrix properties typically reduce PAH bioavailability (Ramesh et al., 2004). Narcosis is the predominant mechanism of toxicity following exposure to complex hydrocarbon mixtures (Tarrad et al., 2024), leading to anaesthesia, hypoactivity, sensory depression, and profound behavioural and physiological alterations in molluscs (Robinson and Rabalais, 2019; MacFarlane et al., 2004). In snails, such exposures impair adhesion, reduce responsiveness, cause tissue damage, promote shell abandonment, and ultimately increase mortality (Farid et al., 2010).

The responses of *Tritia neritea* exposed to diesel-contaminated sediments are consistent with a narcotic mode of action, characterised by locomotor depression and reduced behavioural activity. Control and DIE-1 individuals spent ~25% of their time foraging, whereas DIE-2 snails showed a 50% reduction and remained inactive for prolonged periods (30%), often exhibiting uncoordinated movements. Similar locomotor suppression has been reported in Australian gastropods

Table 5

Summary of behavioural and biochemical responses of *Tritia neritea* under different diesel treatments.

End-point	Response	Key Consequences
Survival	Reduced	Lower survival probability in DIE-2 group.
Burrowing activity	Increased	DIE-2 snails exhibited higher burrowing success.
Predation susceptibility	No effect	-
Foraging success	Reduced	Only 25% of DIE-2 snails reach the food.
Searching time	Increased	DIE-2 snails took nearly twice as long to locate food.
Feeding time	Reduced	Control snails spent 2x more time feeding than diesel group.
Food intake	Reduced	DIE-2 snails consumed significantly less food
Foraging behaviour	Altered	DIE-2 snails showed increased periods of immobility.
Prey selection	Altered	Delayed diet broadening in diesel snails.
Substrate selection	No effect	-
Biochemical Responses	No effect	-

exposed to crude-oil WAF (MacFarlane et al., 2004), and benzo [a] pyrene exposure delayed feeding in *Parhyale hawaiensis* (Lawan et al., 2024). Diesel has also reduced locomotion in *Polinices inaei* at concentrations >75 µg/g (Chapman et al., 1988) and caused disorientation in *Cerithidea fluvialilis* at lower concentrations (Nagarajah et al., 1985). Freshwater snails such as *Bellamya aeruginosa* become inactive under ethylbenzene exposure (Zheng and Zhou, 2017). Excess mucus production has been documented in *T. neritea* and *Nassarius mutabilis* under high PAH exposure (Leung et al., 2020), although this was not observed here. Other stressors such as acidification also impair locomotor and sensory performance (Domenici et al., 2017; Horwitz et al., 2020).

Reduced motility has severe implications for foraging. Snails exposed to the highest diesel concentration required almost twice as long to locate food, despite its proximity, leading to markedly lower foraging success (>75% in Control and DIE-1 vs < 25% in DIE-2). Foraging disruption is a common response to environmental stress: *Nassarius festivus* exhibits reduced locomotion and consumption under low pH (Leung et al., 2015); exposure to Cr and Zn reduces carrion detection (Cheung et al., 2002), elevated pCO₂ impairs foraging in *Reishia clavigera* (Li et al., 2020); and salinity stress reduces activity in *B. atramentaria* and *N. festivus* (Ho et al., 2019; Cheung, 1997). However, beyond external stressors, species-specific traits also play a fundamental role in shaping these toxicological and behavioural outcomes (Cai and Wang, 2019). Furthermore, previous exposure history can modulate uptake and efflux dynamics (Cai and Wang, 2024), which may further contextualize the sensitivity of *T. neritea* observed here. In this sense, the consistent behavioural impairments reported in *T. neritea*, including increased time to locate food under lithium exposure (Marín-Rodríguez et al., 2021), reinforce its potential as a reliable indicator of sediment quality and chemical stress.

Chemically mediated feeding in mud snails involves (i) olfactory detection of carrion and (ii) proboscis extension and ingestion. Distance chemoreception is rapidly inhibited by oil (~48 h), whereas contact chemoreception is affected only at longer exposures or higher WAF concentrations (Hyland and Miller, 1979). In this study, the highest diesel concentration impaired both steps, reducing foraging success and suppressing feeding. Similarly, naphthalene inhibited filtration in the ascidian *Styela plicata* after 1 h (El Ayari et al., 2023). Stickle et al. (1984) reported that the LC₅₀ of *Nucella lima* decreased over 28 days of crude-oil WAF exposure, and that snails exposed to crude-oil WAF concentrations comparable to our DIE-2 (1065 mg/L) ceased feeding entirely, suggesting a common toxicity threshold for petroleum hydrocarbons.

After 7 days, control snails spent twice as long feeding as diesel-exposed individuals, but differences between Control and DIE-1 disappeared by days 14–28, likely due to rapid sediment TPH decline and the reversibility of acute narcosis (Van Wezel and Opperhuizen, 1995). Recovery is well documented: copepods fully recover after transfer to clean media (Saiz et al., 2009), and physiological functions in *Anadara granosa* normalise following naphthalene depuration (Patel and Eapen, 1989). Recovery potential depends on exposure intensity and organismal condition (Ashauer et al., 2016), which can be influenced by feeding before and after exposure (Steigerwald et al., 2025).

DIE-2 snails maintained low feeding activity (<106 s feeding time), and only a few individuals successfully located or consumed prey. This strongly suggests that feeding suppression constrained recovery despite PAH decline. *T. neritea* can accumulate hydrocarbons in the digestive gland, causing degeneration of digestive cells (Sharaf and Shehata, 2015), which may impair feeding. Ineffective feeding response may also reflect reduced locomotion and chemosensory capacity, as observed in other stressed gastropods (Aloy et al., 2011; Chatzinikolaou et al., 2019; Domenici et al., 2017; Guo et al., 2013; Leung et al., 2015; Seuront, 2018). Although glycogen reserves did not differ significantly among treatments, sustained feeding suppression coincided with reduced survival in DIE-2 snails (79 ± 7.5%). Mortality is widely reported in molluscs exposed to petroleum pollutants (Stekoll et al., 1980; Chapman

et al., 1988; de Santana et al., 2021; Monteiro et al., 2018), and PAH bioaccumulation is well documented (Chapman and Connell, 1986; Noh et al., 2018; Silva et al., 2024). The co-occurrence of reduced feeding and higher mortality raises the question of whether mortality reflects direct diesel toxicity or secondary starvation due to behavioural impairment.

Despite reduced feeding and altered locomotion, *T. neritea* consistently preferred *Solen marginatus* when it succeeded in locating prey, not only in DIE-2 but across all treatments. While de Souza et al. (2025) observed shifts toward lower-energy foods in the crab *Eriphia gonagra* after an oil spill, *T. neritea*—both in this study and in Marín Rodríguez et al. (2021)—continued to favour razor clams. This preference aligns with natural dietary patterns (García-García et al., 2015) and the higher energy return provided by *S. marginatus* compared to *Mytilus galloprovincialis* (Marín Rodríguez et al., 2021). Marine invertebrates often select higher-nutrient foods as a compensatory under physiological stress (Sokolova et al., 2012), suggesting that trophic selection in *T. neritea* may represent an adaptive strategy, particularly under high diesel exposure.

Regarding righting success, although differences were not statistically significant, DIE-2 individuals showed poorer performance. The righting response is a critical anti-predator behaviour that reduces the predation risk but is sensitive to environmental stress. Delayed righting has been documented under low pH (*Titra obsolete* and *Tegula funebris*, Froehlich and Lord, 2020; Jellison et al., 2016), lithium exposure (*T. neritea*, Marín-Rodríguez et al., 2021), venlafaxine (*Ilyanassa obsoleta* Fong et al., 2017), fluoxetine (*Gibbula unilicalis*, Ford et al., 2018), and other antidepressants (*Ilyanassa obsoleta* and *Leptoxis carinata*, Fong et al., 2019). Because anti-predator responses often come at the expense of foraging, a trade-off is expected: stronger predator responses when food is abundant or body condition is high (Büchner-Miranda et al., 2024).

Burrowing is another essential activity in mud snails. DIE-2 snails showed a greater tendency to burrow, but diesel exposure produced only minor differences in substrate selection or subsequent burrowing. Most studies report reduced burrowing under hydrocarbon exposure in snails (*Polinices* spp.; Gulec et al., 1997; Chapman et al., 1988), bivalves (*Anadara granosa*-Patel and Eapen, 1989), *Macoma balthica*-Taylor and Karinen (1977), fiddler crabs (*Uca longisignalis*, *U. panacea* Karam and Al-Wazzan, 2021), and polychaetes (*Hediste diversicolor*; Bagwell, 2020). Some species avoid settling or burrowing in contaminated sediments (Araújo et al., 2012; Karam and Al-Wazzan, 2021). Chemosensory detection may allow organisms to detect pollutants and trigger avoidance (Dell’Omo, 2002; Amiard-Triquet and Amiard, 2012). This was not observed here, supporting the hypothesis that behavioural alterations in DIE-2 snails may arise from impaired detection of chemical cues, similar to effects in *T. obsoleta* and other snails under ocean acidification (Froehlich and Lord, 2020; Jacobson and Boylan, 1973; Jellison et al., 2016; Manz et al., 2023). PAHs may interfere with receptor–ligand binding (e.g., GABA receptors; Olasehinde and Olaniran, 2022) or with signal transmission to the olfactory lobe (Filatova and Abramochkin, 2023). If snails fail to perceive threat cues, they may reduce movement or increase burrowing until conditions improve.

Behavioural alterations may also arise from underlying biochemical detoxification and cellular defence processes, which demand substantial energetic investment. Because these mechanisms divert resources away from growth, reproduction, and general activity, organisms may compensate by reducing energetically costly behaviours such as feeding and burrowing (Dell’Omo, 2002). Thus, the behavioural impairments observed here may result from metabolic adaptations to maintain homeostasis under chemical stress. This interpretation is consistent with the observed reduction in GLY, suggesting enhanced use of internal energy reserves. However, in the present study, diesel exposure did not elicit measurable biochemical responses in SOD, and LPO. The absence of enzymatic activation or inhibition suggests that, at the tested concentrations, diesel either (i) did not reach toxicity thresholds necessary

to trigger classical antioxidant or (ii) induced effects primarily at higher organisational levels before biochemical disruption became detectable. Nevertheless, signs of detoxification activity were evident at the lowest diesel concentration, suggesting that organisms may be able to mount a defence response only under moderate exposure levels. The absence of similar responses at the highest concentration may reflect a threshold beyond which detoxification capacity becomes overwhelmed or suppressed. Van der Oost et al. (2003) demonstrated that detoxification and antioxidant enzymes frequently follow a biphasic pattern, increasing at low pollutant concentrations but diminishing when exposure exceeds physiological capacity. Also, Regoli and Giuliani (2014) showed that organisms can activate detoxification pathways at lower contaminant levels, but high concentrations can inhibit or deplete these systems, leading to reduced biomarker responses. Neurotoxicity associated with the exposure and bioaccumulation of petroleum-derived compounds can also shape behavioural outcomes (Basu, 2015). The literature available demonstrates that gastropods exhibit highly variable biochemical responses to hydrocarbons, and a lack of enzymatic change does not exclude stress. The literature available demonstrates that gastropods exhibit highly variable biochemical responses to hydrocarbons, and a lack of enzymatic change does not exclude stress. For example, diesel increased antioxidant defences in *Nacella concinna* (Ansaldi et al., 2005), while water accommodated fractions (WAF) of crude oil triggered oxidative stress responses in *Neritina virginea* (Sandrini-Neto et al., 2016) and *N. concinna* (Feijó de Oliveira et al., 2015). Conversely, WAF exposure in *Pomacea patula* elevated reactive oxygen species (ROS) without altering SOD, catalase (CAT) or Glutathione Peroxidase (GPx) activities (Olivares-Rubio et al., 2017), illustrating that oxidative imbalance can occur without canonical biomarker modulation. In other cases, even chronic exposure scenarios—such as recurrent oil spills—failed to induce enzymatic defences in *N. virginea* (Sandrini-Neto et al., 2016).

In the present study, the inhibition of AChE activity at the highest diesel concentration provides further support for this mechanism, as reduced cholinesterase activity can impair synaptic transmission and motor coordination. Such neurophysiological disruption is consistent with the diminished feeding and burrowing behaviours observed, suggesting that at elevated contamination levels, diesel not only challenges metabolic and detoxification processes but also compromises neuromuscular function, thereby amplifying behavioural impairments. Gai-tonde et al. (2006) measured AChE activity in the marine snail *Cronia contracta* along the Goa coast and found reduced AChE associated with elevated petroleum hydrocarbons and neurotoxic metals in tissues, supporting the use of AChE inhibition as a biomarker of neurotoxic contamination. Also, Zheng & Zhou (2017) demonstrated that AChE was inhibited in the freshwater snail *Bellamya aeruginosa* exposed to PAHs. Xuereb et al. (2009) explicitly linked AChE inhibition with ecologically relevant behavioural endpoints (feeding, locomotion) in the amphipod *Gammarus fossarum*, showing that reduced cholinesterase activity impairs normal movement and feeding behaviour.

5. Conclusion

Taken together, these findings demonstrate that gastropod biochemical pathways do not respond uniformly to petroleum hydrocarbons, reinforcing the view that behavioural endpoints may serve as more sensitive and earlier indicators of sublethal stress. The behavioural impairments observed in the absence of consistent biomarker responses suggest that diesel may exert its effects through mechanisms not fully captured by the biochemical assays applied here, such as subtle neurophysiological disruption, altered energy allocation, or sensory impairment. This mismatch between organism-level behaviour and cellular-level biomarkers underscores the complexity of petroleum-derived toxicity and highlights the limitations of relying exclusively on enzymatic responses.

Even subtle behavioural changes may have ecologically relevant

consequences for mud snails, which often form dense populations and constitute an important food resource for multiple predators (Zhang et al., 2021; Uehara et al., 2024). Increased burrowing activity may reduce prey availability, potentially altering predator foraging behaviour (Tucker et al., 1997). Likewise, reduced escape responses in diesel-exposed snails could modify predator–prey dynamics leading to shift in food-web structure (Froehlich and Lord, 2020). Such effects may propagate through bottom-up processes, whereby reduced prey accessibility or consumption contributes to predator declines and subsequent alterations in ecosystem function. In addition, mud snails play a key role in nutrient recycling in mudflats, sandflats, and salt marshes (Zhang et al., 2021). Under hydrocarbon contamination, increased burrowing and reduced locomotion may lower foraging rates, thereby decreasing nutrient turnover. Reduced grazing on algae and carrion may further exacerbate low-oxygen conditions by enhancing organic matter accumulation and decomposition in shallow-water habitats.

Overall, these results emphasise the importance of integrative, multi-level ecotoxicological approaches that combine behavioural, biochemical, and physiological endpoints. Behavioural disruptions, which often precede detectable biochemical changes, have the potential to scale up to higher levels of biological organization, ultimately leading to ecologically meaningful consequences.

CRedit authorship contribution statement

Mercedes Conradi: Writing – review & editing, Formal analysis, Conceptualization. **María Maguilla:** Writing – review & editing, Formal analysis. **J. Emilio Sánchez-Moyano:** Writing – review & editing, Formal analysis. **Marta Cunha:** Writing – review & editing. **Camilo Dias Seabra Pereira:** Writing – review & editing. **Rosa Freitas:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marenvres.2026.107948>.

Data availability

Data will be made available on request.

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